

$$\begin{array}{c}
 M_L \nearrow M_H \\
 (V_L, U_L, R_L(V_L), F_L) \quad (V_H, U_H, R_H(V_H), F_H)
 \end{array}$$

No "abstraction" $\implies V_L = V_H$.

• approximate abstraction distance
 d_T = "worst case error" distance
 $d_{\max}$.

$\tau: R(V_L) \rightarrow R(V_H)$ [state mapping]

$$d_E(M_L, M_H) = \min_{\{T_U: T_U(P_L) = P_H\}} \max_{\{\vec{x} \leftarrow \vec{x} \in I\}}$$

Expectation $\left(\sum_{\vec{u}_L \in R_L(U_L)} Pr_L(\vec{u}_L) \underbrace{d_{V_H}(\tau(M_L(\vec{u}_L), \vec{x} \leftarrow \vec{x})), M_H(\tau_U(\vec{u}_L), w_T(\vec{x} \leftarrow \vec{x})))}_{\text{Difference}} \right)$

$$\tau(Pr_L \vec{x} \leftarrow \vec{x}) = Pr_H w_T(\vec{x} \leftarrow \vec{x}) \quad \checkmark$$

$\min T_U \rightarrow$ best case

$\max \vec{x} \leftarrow \vec{x} \rightarrow$ worst case Intervention.

$$\tau: R(V_L) \rightarrow R(V_H)$$

Intervention distrib. Pr_I is a distrib. on $I_L \times I_H$ s.t

$$Pr_I(\vec{X} \leftarrow \vec{x} | \vec{Y} \leftarrow \vec{y}) > 0 \text{ iff}$$

$$w_I(\vec{X} \leftarrow \vec{x}) = \vec{Y} \leftarrow \vec{y}$$

lifting this
low level
intervention

gives us this
high level
one

$$d_I^{Pr_I}(M_L, M_H) = \min_{\{T_u: T_u(Pr_L) = Pr_H\}} \max_{\vec{Y} \leftarrow \vec{y} \in I_H}$$

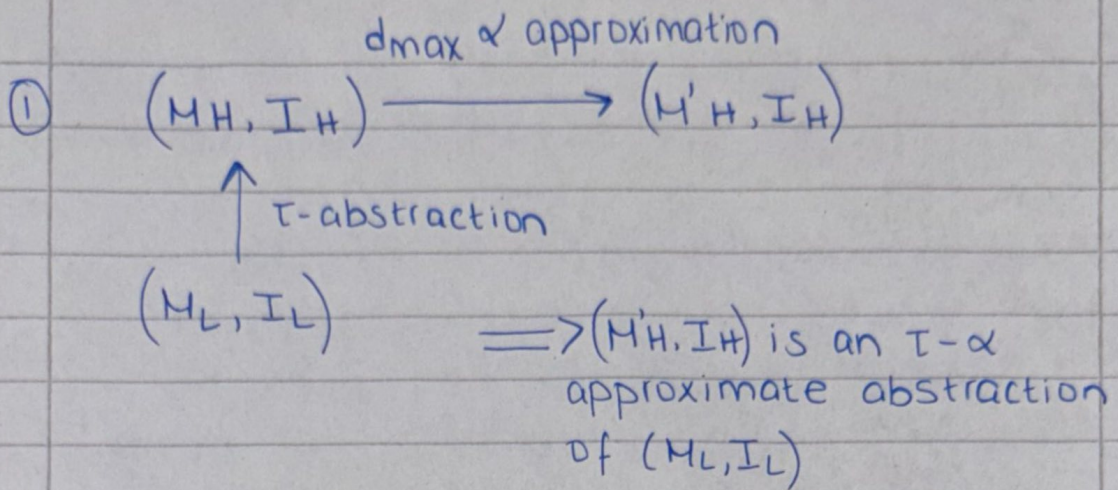
$$\left[\sum_{\vec{u}_L \in R_L(\vec{u}_L)} Pr_L(\vec{u}_L) \cdot \left(\sum_{\vec{X} \leftarrow \vec{x} \in I_L} Pr_I(\vec{X} \leftarrow \vec{x} | \vec{Y} \leftarrow \vec{y}) \right) \right] \right]$$

$$d_{V_H}(\tau(M_L(\vec{u}, \vec{X} \leftarrow \vec{x}), M_H(T_u(\vec{u}), w_I(\vec{X} \leftarrow \vec{x})))$$

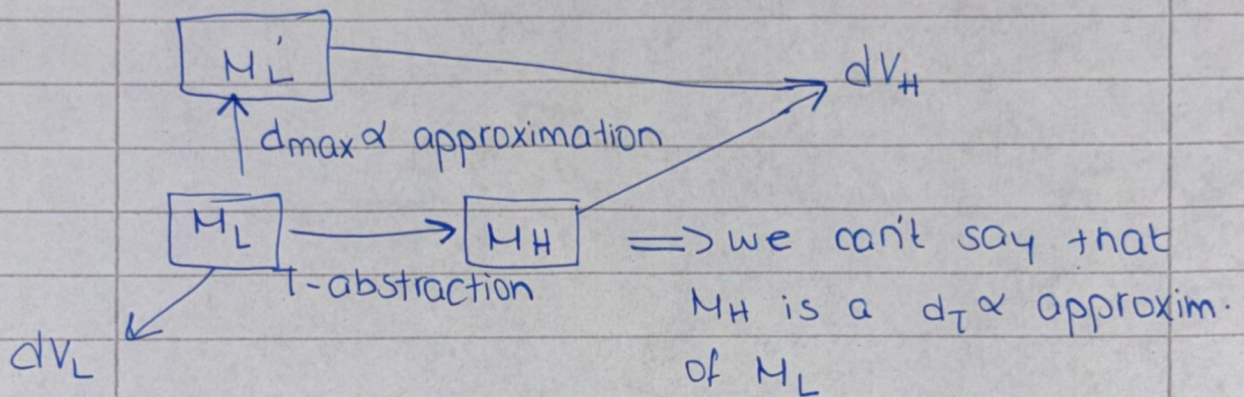
unchanged, same formula

① averages error across all likely low level contexts? weighted by probab. Pr_L

② averages error across all possible low level implementations $(X \leftarrow x)$ of that high level action, weighted by the interventional distrib. Pr_I .



② Opposite order:



Theorem 5.3:

*(M_H is noisy version of M'_H)

